

Forum Module – Technical Requirements and Implementation Guidelines

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1. Advanced Permissions System

The purpose of this requirement is to implement a role-based access control system (RBAC) that determines which actions a user can perform within the forum platform.

The system should support:

1. managing user roles,
2. assigning roles to users,
3. permissions restrict access to specific features,
4. displaying different UI views depending on the user's role.

Roles:

1. Administrator
2. Moderator
3. User
4. Guest

Permissions:

1. create forums,
2. delete forums,
3. create posts,
4. delete posts,
5. manage users,
6. manage organizations.

2. Organization System (Forums)

The purpose of this requirement is to implement organizations that act as independent forums or communities within the platform.

The system should support:

1. creating organizations,
2. editing organizations,
3. deleting organizations,
4. adding users to organizations,
5. removing users from organizations,
6. managing memberships,
7. allowing users to perform actions inside organizations based on their permissions.

Example:

Organization:

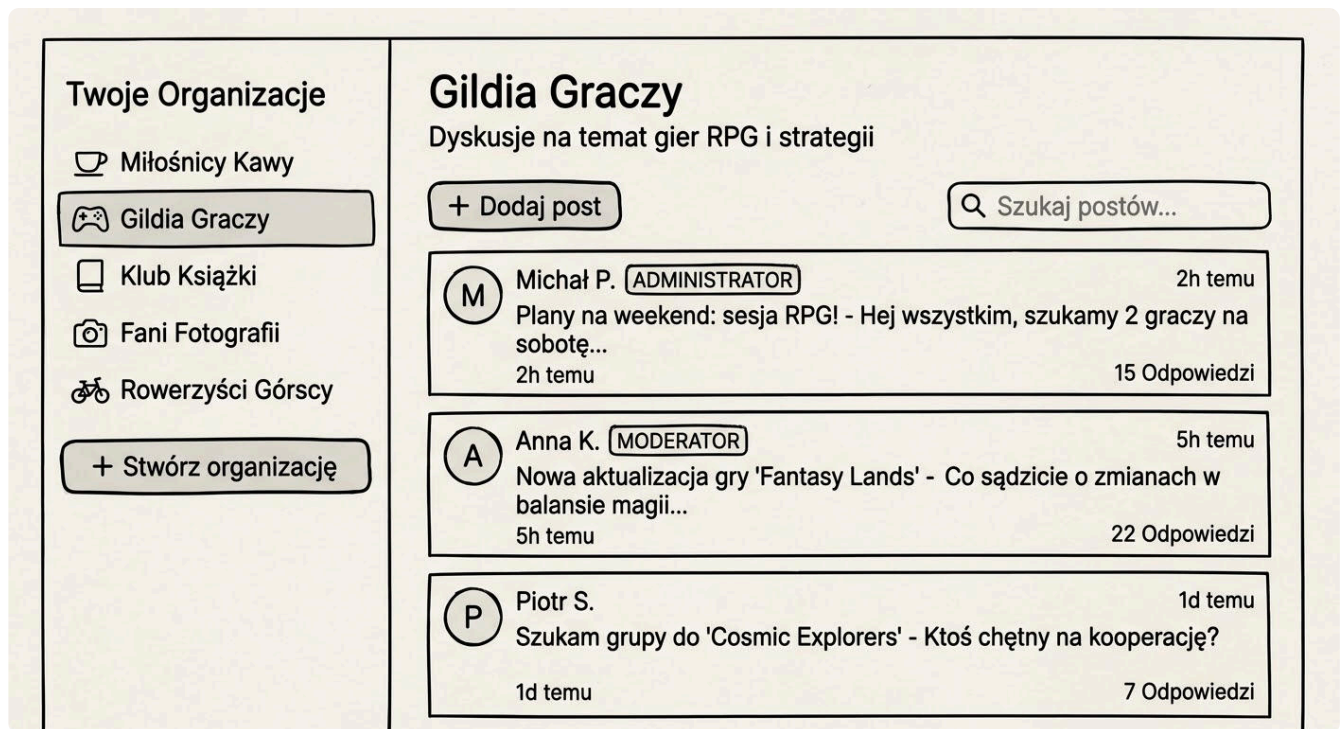
* “42 Warsaw”

Members:

1. Organization Admin
2. Organization Moderator
3. Organization Member

Each organization may contain its own:

1. categories,
2. posts,
3. members,
4. permission settings.



Sketch

3. Forum Main View

The Forum Main View serves as the entry point for users to discover and join organizations within the platform.

Search Bar

A real-time search bar allowing users to search for existing organizations by name. The search results are dynamically updated as the user types.

Example:

- Typing “dog” will display organizations such as “Dog Lovers”.
- Typing “gaming” will display organizations related to gaming.

Create Organization Button

A dedicated button that redirects users to the organization creation form, where they can create a new organization and configure its basic settings.

Recommended Organizations

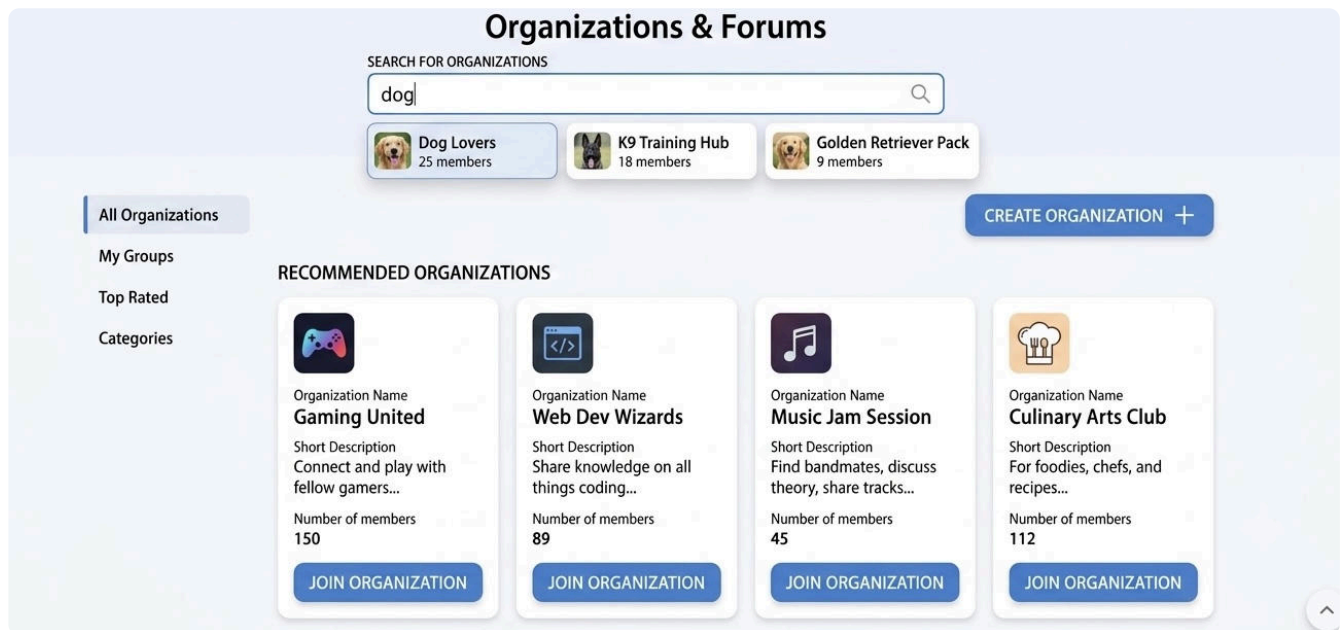
The homepage displays a list of recommended organizations that users may be interested in joining. These recommendations can be based on popularity, activity, or predefined selections.

Each organization card should display:

1. Organization name
2. Short description

3. Number of members
4. Organization avatar/logo (optional)
5. “Join Organization” button

Users can join an organization directly from the main forum page without needing to navigate to a separate view.



Example

4. Organization Creation Form

The Organization Creation Form allows users to create a new organization (forum) within the platform.

Access

Users can access the form by clicking the “Create Organization” button available on the Forum Main View.

1. Required Fields
2. Organization Name
3. A unique name identifying the organization.

Requirements:

1. Must be unique across the platform.
2. Cannot be empty.

3. Must meet the application's validation rules regarding length and allowed characters.

Member Limit

The maximum number of members allowed to join the organization.

Requirements:

1. *Must be a positive integer.
2. Defines the maximum capacity of the organization.
3. Example: 5 members, 20 members, 100 members.

Description

A short description explaining the purpose and goals of the organization.

Requirements:

1. Cannot be empty.
2. Should provide enough information for users to understand the organization's purpose.
3. Maximum character limit to be defined by the development team.

Organization Creation Process

When the user submits the form:

1. The system validates all provided data.
2. The system verifies that the organization name is unique.
3. A new organization record is created in the database.
4. The creator is automatically added as a member of the organization.
5. The creator is automatically assigned the Organization Administrator role.
6. The organization becomes visible in the Forum Main View and searchable through the organization search system.

Organization Administrator Permissions

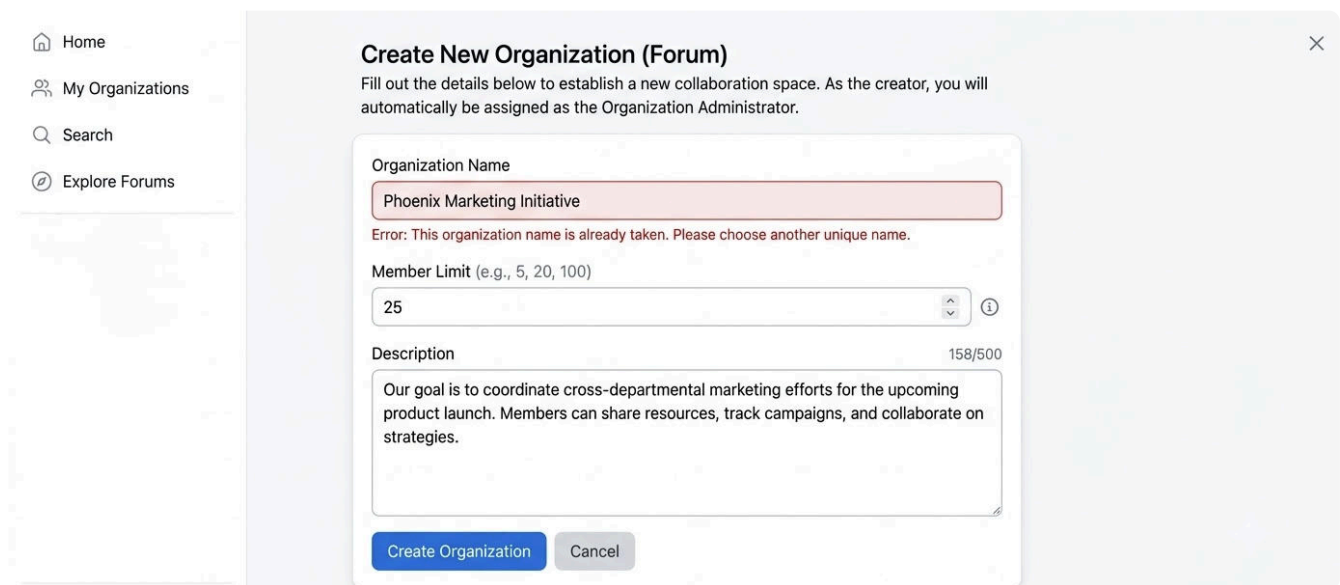
The Organization Administrator is granted the highest level of permissions within the organization and can:

1. Edit organization information.
2. Delete the organization.
3. Invite or add users to the organization.
4. Remove users from the organization.
5. Manage organization membership.

6. Assign organization roles (if role management is implemented).
7. Perform all actions available to regular organization members.

Error Handling

1. The system should display appropriate validation messages when:
2. The organization name already exists.
3. Required fields are missing.
4. The member limit contains an invalid value.
5. The organization cannot be created due to a server or database error.



The screenshot shows a web interface for creating a new organization. On the left is a sidebar with links: Home, My Organizations, Search, and Explore Forums. The main content area is titled 'Create New Organization (Forum)' and includes a sub-header: 'Fill out the details below to establish a new collaboration space. As the creator, you will automatically be assigned as the Organization Administrator.' The form contains three fields: 'Organization Name' with the value 'Phoenix Marketing Initiative' and a red error message 'Error: This organization name is already taken. Please choose another unique name.'; 'Member Limit (e.g., 5, 20, 100)' with a value of 25; and 'Description' with the text 'Our goal is to coordinate cross-departmental marketing efforts for the upcoming product launch. Members can share resources, track campaigns, and collaborate on strategies.' At the bottom are 'Create Organization' and 'Cancel' buttons.

Example

5. Organization Roles and Permission Management

Each organization must maintain its own role hierarchy and permission system. Roles are assigned per organization, meaning a user can have different roles in different organizations.

Example:

- * User A can be an Administrator in Organization X.
- * The same User A can be a regular Member in Organization Y.

□

Roles Management View

Each organization must contain a dedicated Roles & Members section accessible from the organization dashboard.

This view allows members to:

1. View all organization members.
2. View the role assigned to each member.
3. See the organization's role hierarchy.
4. Manage roles (depending on permissions).

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- * View all organization members.
- * View the role assigned to each member.
- * See the organization's role hierarchy.
- * Manage roles (depending on permissions).

Displayed information:

User	Role
John Doe	Administrator
Alice Smith	Moderator
Bob Wilson	Member

Available actions depend on the current user's role.

□

6 . Role Assignment Rules

Organization Administrator

The creator of the organization automatically receives the Organization Administrator role during organization creation.

There must always be at least one Administrator in an organization.

Administrator permissions:

- * Edit organization name.
- * Edit organization description.
- * Edit member limit.
- * Delete the organization.
- * View all members.
- * Add users to the organization.
- * Remove users from the organization.
- * Promote members to Moderator.
- * Demote Moderators to Member.
- * Transfer administrative privileges to another member (optional future feature).
- * Delete any post within the organization.
- * Access all organization management features.

Administrator restrictions:

- * Cannot remove themselves if they are the last Administrator.
- * Cannot leave the organization without transferring ownership or deleting the organization.

□

Moderator

Moderators assist Administrators in managing the organization.

A Moderator role can only be assigned by an Administrator.

Moderator permissions:

- * View all organization members.
- * Remove regular Members from the organization.
- * Delete posts created by Members.
- * Moderate organization content.

- * Report inappropriate content.

Moderator restrictions:

- * Cannot modify organization settings.
- * Cannot delete the organization.
- * Cannot assign or remove Administrator roles.
- * Cannot remove Administrators.
- * Cannot promote users to Administrator.

□

Member

Members are standard participants of the organization.

The Member role is automatically assigned when a user joins an organization.

Member permissions:

- * View organization content.
- * Create posts.
- * Edit their own posts.
- * Delete their own posts.
- * Comment on posts.
- * Participate in discussions.

Member restrictions:

- * Cannot manage organization settings.
- * Cannot assign roles.
- * Cannot remove users.
- * Cannot edit or delete content created by other members.

□

Guest

Guests are users who are not members of the organization.

Guest permissions:

- * View public organizations.

- * View organization descriptions.

- * View public content (if enabled).

Guest restrictions:

- * Cannot create posts.

- * Cannot comment.

- * Cannot join private organizations without invitation.

- * Cannot access member-only content.

- * Cannot manage any organization resources.

□

Permission Matrix

Action	Administrator	Moderator	Member	Guest
View Organization	✓	✓	✓	✓ (Public Only)
Create Post	✓	✓	✓	✗
Edit Own Post	✓	✓	✓	✗
Delete Own Post	✓	✓	✓	✗
Delete Any Post	✓	✓	✗	✗
Add Users	✓	✗	✗	✗
Remove Members	✓	✓	✗	✗
Remove Moderators	✓	✗	✗	✗
Remove Administrators	✓	✗	✗	✗
Assign Moderator Role	✓	✗	✗	✗
Edit Organization	✓	✗	✗	✗
Delete Organization	✓	✗	✗	✗
View Members List	✓	✓	✓	✗

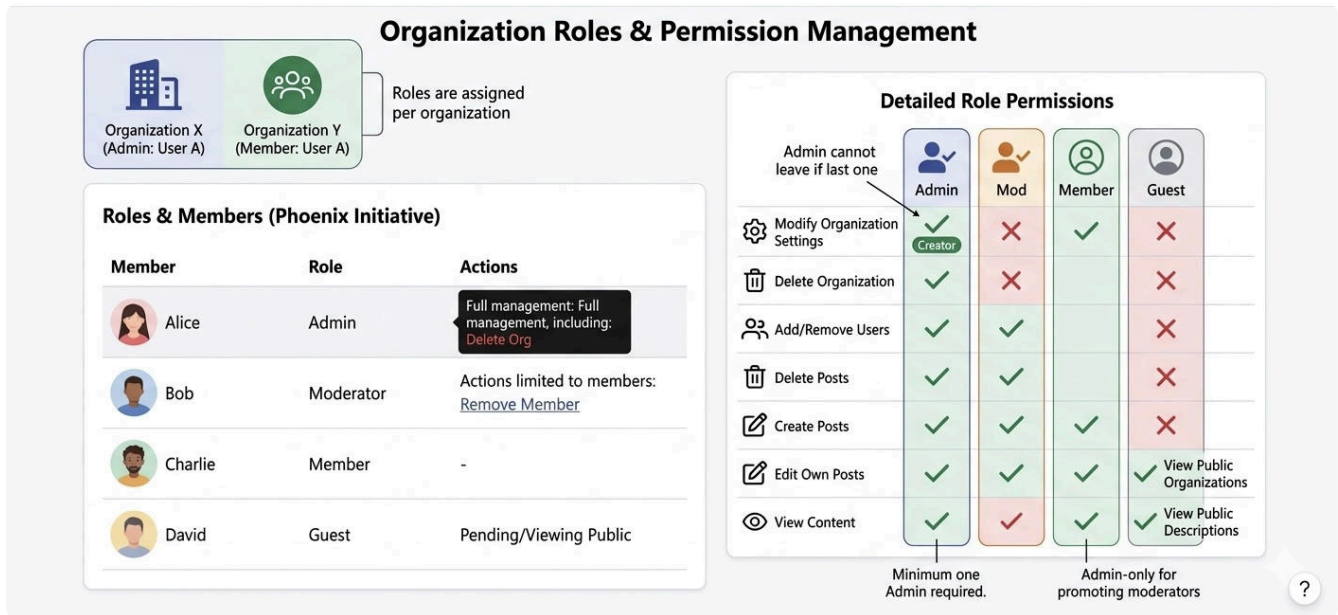
Access Control Requirements

All permissions must be enforced on both:

- * Frontend (UI visibility and available actions)

- * Backend (API authorization and validation)

The backend must always validate the user's role before executing any protected action to prevent unauthorized access through direct API requests.



Example

7 . Organization Posts and Content Management

Posts are the primary form of communication within an organization. Members can create, view, and interact with content according to their assigned role.

□

Post Visibility Rules

The visibility of posts depends on the organization's access level.

Public Organization

Posts can be viewed by:

- * Administrator
- * Moderator
- * Member
- * Guest (read-only)

Guests cannot interact with posts.

□

Private Organization

Posts can only be viewed by:

- * Administrator

- * Moderator

- * Member

Guests cannot access any organization content.

□

Creating Posts

Only organization members can create posts.

Eligible roles:

- * Administrator

- * Moderator

- * Member

Guests cannot create posts.

□

Create Post Form

A post creation form should contain the following fields:

Title

A short title describing the topic of the post.

Requirements:

- * Required field.

- * Maximum length defined by the development team.

Content

Main text content of the post.

Requirements:

- * Required field.

- * Supports multiline text.

Attachments (Optional)

Users may attach files to a post.

Supported attachment types:

- * Images (.jpg, .jpeg, .png, .webp)

- * Documents (.pdf)

Requirements:

- * Maximum file size configurable by administrators.

- * Multiple attachments may be supported.

Publish Button

Creates the post and makes it visible to authorized users.

□

Post Ownership

Each post must store:

- * Author

- * Creation date

- * Last modification date

- * Organization identifier

The author automatically becomes the owner of the post.

□

Editing Posts

Administrator

Can edit:

- * Any post.

Moderator

Can edit:

- * Any post.

Member

Can edit:

- * Only their own posts.

Guest

Cannot edit posts.

□

Deleting Posts

Administrator

Can delete:

- * Any post.

Moderator

Can delete:

- * Any post.

Member

Can delete:

- * Only their own posts.

Guest

Cannot delete posts.

□

Comments System

Users can interact with posts through comments.

Who Can Comment

Role Can Comment

Administrator ✓

Moderator ✓

Member ✓

Guest ✗

□

Creating Comments

Comment form:

* Comment text

* Submit button

Requirements:

- * Comment text cannot be empty.

- * Author information is stored.

- * Creation date is stored.

□

Editing Comments

Administrator

Can edit:

- * Any comment.

Moderator

Can edit:

- * Any comment.

Member

Can edit:

- * Only their own comments.

Guest

Cannot edit comments.

□

Deleting Comments

Administrator

Can delete:

- * Any comment.

Moderator

Can delete:

- * Any comment.

Member

Can delete:

- * Only their own comments.

Guest

Cannot delete comments.

□

Content Moderation

To maintain order within organizations:

Administrators and Moderators can:

- * Remove inappropriate posts.
- * Remove inappropriate comments.
- * Review reported content.
- * Take moderation actions when community rules are violated.

□

Organization Posts and Content Management

Organization Posts - Public View

Phoenix Initiative

New Marketing Plan Draft
Author: Alice (Admin)
Date: 2026-06-02
Last Edited: 2026-06-03

Interaction Icons Reply Likes

Feedback on Logo
Author: Charlie (Member)

Visibility Rules & Interactions Matrix

Org Type	View Posts	Interact (Like/Reply)	Create Post	Edit (Own/All)	Delete (Own/All)
Public / Private	🌐	♥	✓	✎	✗
Guest	✓	View-only	✓	✓	✗
Guest / Member	✓	✓	✓	✓	✗
Moderator	✓	✓	✓	✓	✗
Admin	✓	✓	✓	✓	Moderate

Create New Post Form

Alice Admin

Title

Content

Attachments (Optional)

phoenix_logo_v2.png 1.85 MB

marketing_strategy.pdf 22.05 KB

Publish Post

Content Moderation Queue

Only listed if the user is logged in as an Admin or Moderator.

Post or Comment

Comment: This is spam
Author: GuestX
Reported By: Charlie (Member)

Action Delete Dismiss Review

Moderation Status Pending Review

Sketch

8. Membership Management (Join & Leave Flow)

This module controls the flow of users within organizations.

Joining (Join):

User sends a request to join the organization.

Validation: The system checks if the current member count is below the defined limit (e.g., the 5-person limit you mentioned).

Pending State: If the organization is restricted, the request appears in the Admin's "Pending Requests" tab.

Leaving (Leave):

Users can leave at any time.

Exception: If the sole Administrator tries to leave, the system must force them to transfer the role to another member or delete the organization.

2. Advanced Permissions System (Role Assignment Flow)

Following the "Advanced permissions system" requirement, every action on both frontend and backend must be verified against the user's role.

Role Hierarchy & Capabilities:

1 Administrator (Owner):

Full CRUD (Create, Read, Update, Delete) permissions at the organization level.

Role Management: Can upgrade a standard User to a Moderator.

Access Control: Can kick any member from the group.

Settings: Can modify the description and the member limit.

2 Moderator:

A role assigned by the Administrator.

Has content moderation rights (can delete other people's posts if they violate rules).

Can remove standard Users from the organization (but cannot kick the Admin).

3 User:

Can read posts, create new ones, and edit/delete only their own content.

4 Guest:

Read-only access to public organization information.

Role Assignment Process (Step-by-Step):

Step 1: The Admin opens the member list of the specific organization.

Step 2: Selects a specific user.

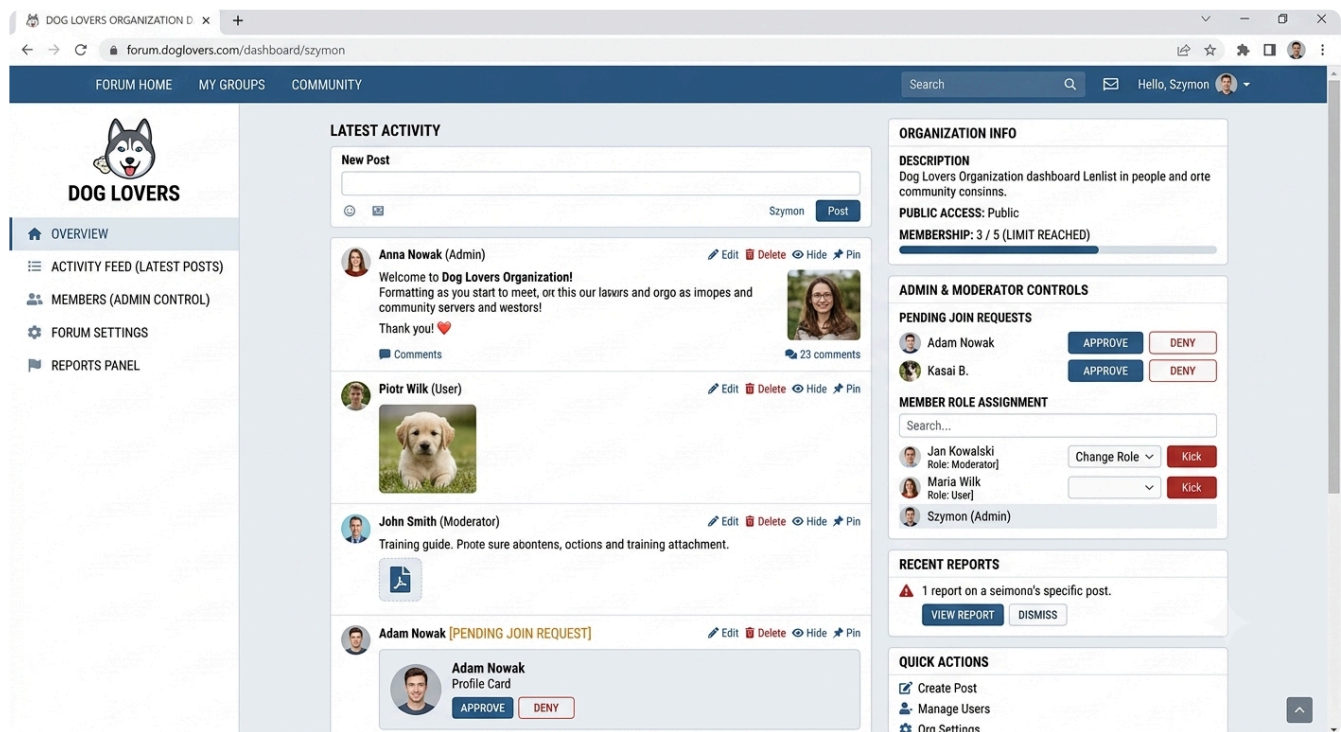
Step 3: Clicks "Change Role" and selects from the dropdown: "Moderator" or "User".

Step 4 (Validation): The Backend verifies:

1 Does the requester have the is_admin flag in this organization?

2 Is the target not the Admin themselves (self-demotion is blocked without power transfer)?

Step 5: Database record update and permission refresh (e.g., via WebSockets or state update) without requiring a full page reload.



9. Forum Management and Organization Workflow

Forum Discovery Page

The Forum page is the entry point to the module.

Displayed sections:

Available Organizations

Users can browse existing organizations.

Each organization card displays:

- * Organization name
- * Short description
- * Current member count

* Visibility type:

* Public

* Private

Available actions

* Join Organization

* View Organization (if permitted)

Create Organization

Authorized users can create a new organization.

Organization creation form:

* Organization Name

* Description

* Visibility Type

* Public

* Private

* Member Limit

Upon creation:

* The creator automatically becomes the Administrator.

* The organization becomes available in the forum list.

□

Organization Workspace

After entering an organization, users are redirected to the Organization Workspace.

The workspace contains two primary sections:

Posts

Displays all posts visible to the current user.

Each post card contains:

* Title

* Author

* Creation Date

- * Last Modification Date

- * Number of Comments

Available actions depend on user role.

Users may:

- * Open a post

- * Create a new post

- * Edit a post (if authorized)

- * Delete a post (if authorized)

□

Members

The Members tab displays all organization members.

Displayed information:

- * Username

- * Role

- * Join Date

Available actions depend on permissions.

□

Role Management

Role management is performed directly from the Members tab.

This eliminates the need for a separate administration panel while still fulfilling the Advanced Permissions System requirement.

Administrator

Can:

- * Promote User to Moderator

- * Demote Moderator to User

- * Remove members

- * Approve membership requests

- * Modify organization settings

Restrictions:

Cannot remove themselves without transferring ownership.

Membership Management

Joining an Organization

Step 1

User clicks:

Join Organization

Step 2

Backend validates:

- * User is not already a member.
- * Member limit has not been reached.

Step 3

If the organization is private:

- * A membership request is created.
- * Status = Pending.

Step 4

Administrator can:

- * Accept request.
- * Reject request.

□

Leaving an Organization

Members may leave an organization at any time.

Validation:

If the leaving member is the only Administrator:

The system must require:

- * Ownership transfer

or

- * Organization deletion

before leaving.

□

Organization Settings

Organization settings are accessible only to Administrators.

Editable fields:

- * Organization Name
- * Description
- * Visibility Type
- * Member Limit

All changes are validated by the backend.

□

Content Moderation

Moderation tools are integrated directly into posts and comments.

No separate moderation dashboard is required.

This keeps the workflow simple and aligned with project requirements.

Moderation Actions

Administrators and Moderators can:

- * Remove posts
- * Remove comments
- * Review reported content
- * Enforce organization rules

Moderation buttons are displayed directly next to the content when the user has sufficient permissions.

□

Permissions Validation

Every protected action must be validated on:

Frontend

Used for:

- * Hiding unavailable actions.

- * Improving user experience.

Backend

Used for:

- * Security enforcement.

- * Preventing unauthorized requests.

The backend remains the single source of truth.

□

Real-Time Permission Updates

When a role is changed:

- * Database is updated.

- * Organization permissions are refreshed.

- * User interface updates without page reload.

Possible implementation:

- * WebSockets

- * Server-Sent Events

- * State synchronization after mutation

Navigation Flow

Forum Page

|

|— Browse Organizations

|

|— Create Organization

|

|— Join Organization

|



Organization Workspace



Depending on user role.

10. Final Module Visualization

Overview

The Forum module is designed around a simple and intuitive user experience while maintaining a complete role-based permission system.

The entire workflow is divided into three main views

1. Forum Dashboar
2. Organization Workspace
3. Members Management

The navigation structure remains consistent across the entire platform.

Global Navigation

The following navigation tabs are visible on every page:

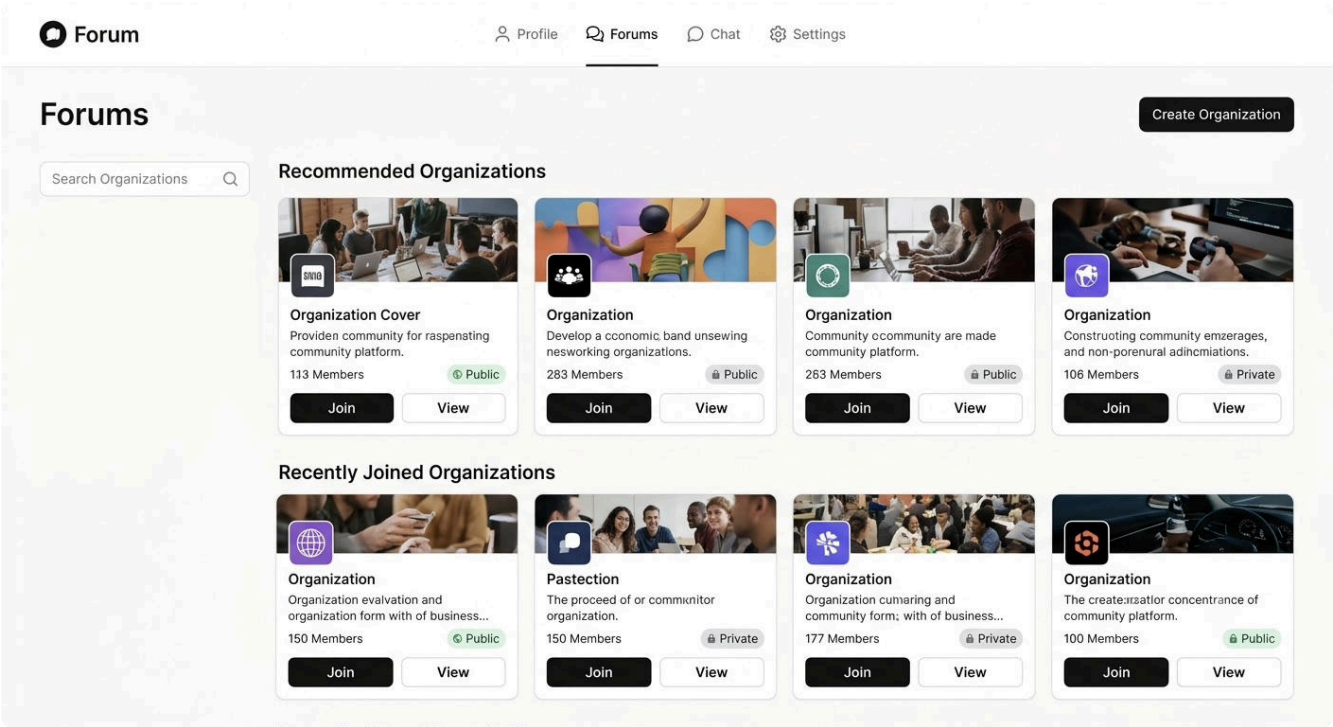
- * Profile
- * Forums
- * Chat
- * Settings

Access to actions and content depends on the user's role within a specific organization.

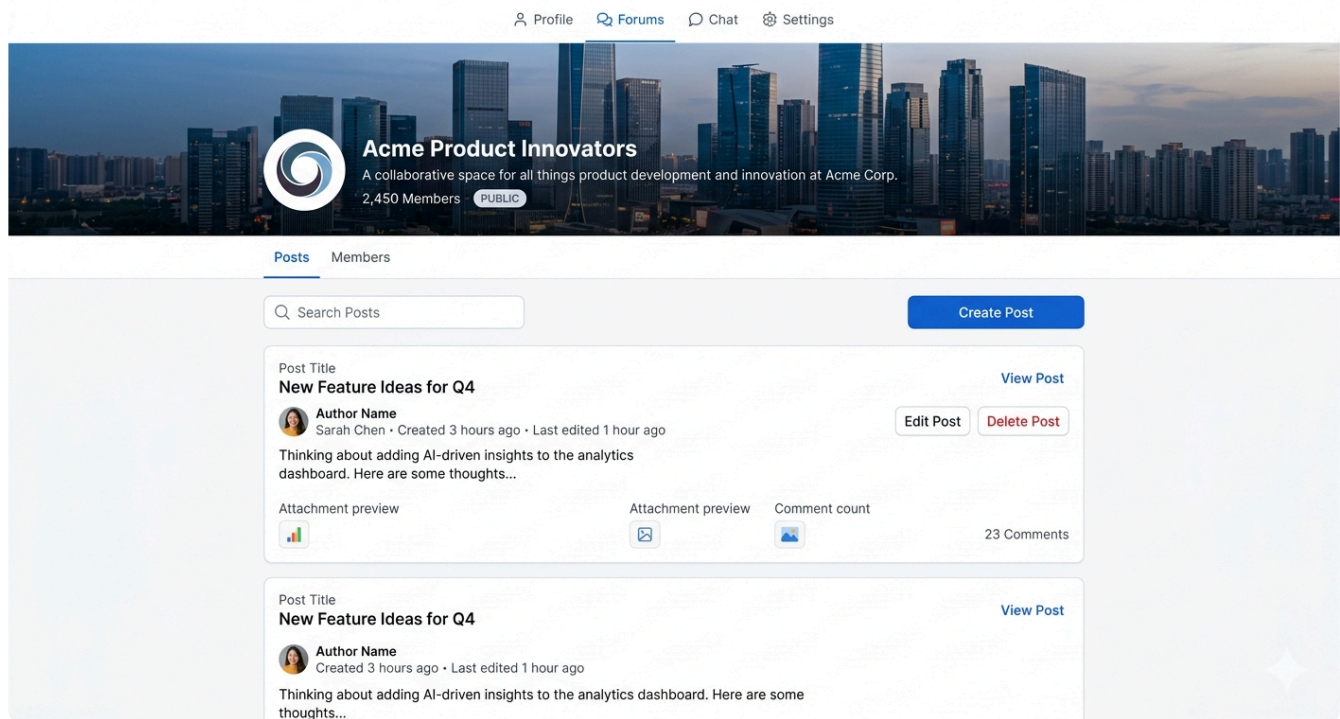
Role-Based Access

The system dynamically adjusts available actions depending on the user’s role.

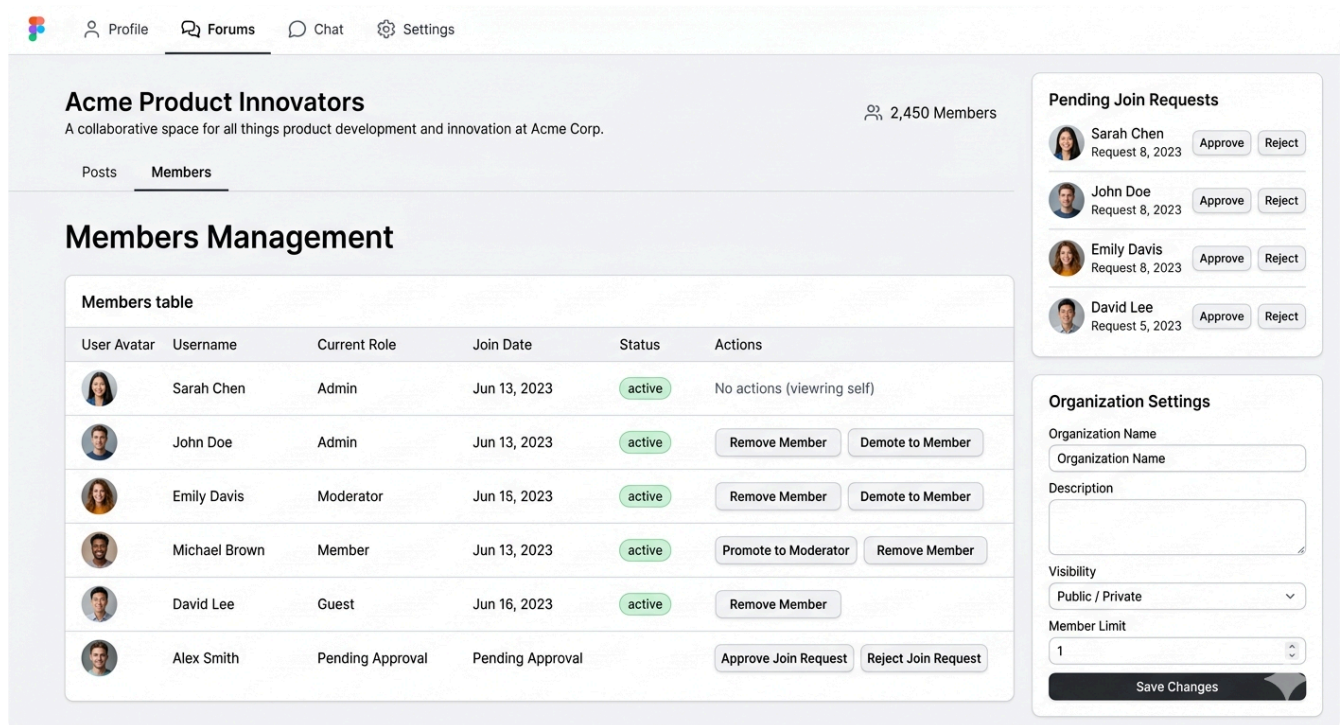
Forum Dashboard → Organization Workspace → Members Management (Admin View)



Forum Dashboard



Organization Workspace



Members Management

The above visualizations are used to illustrate how this module should look like, please focus on what was described in this article, if you don't know something, write to me or read the article again, visualizations are only to be able to imagine something better.